

David Zhao

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EDUCATION

Wilfrid Laurier University

Waterloo, ON

Bachelor of Science in Data Science(co-op) - 4.0 Program GPA

Sept 2024 - Present

- **Relevant Courses:** Data Structures, OOP, Data Visualization, Database, Calculus II, Computer Networks
- **Clubs:** Laurier Analytics Society - VP of Tech

EXPERIENCE

Software Developer Intern

May 2026 – Present

DOUBL

Toronto, ON, Canada

- **Architected** a split dev/prod Firebase environment with isolated Firestore databases and storage buckets, eliminating test data contamination and protecting **1,000+ real user** records from dev-cycle risk, improving development workflow safety
- **Upgraded** Doubl's static website into a full-stack ML-powered platform by integrating **33 responsive, mobile compatible pages** with business-driven features, accelerating the product's readiness for early user onboarding
- **Collaborated** with a senior engineer and product owner through daily standups and weekly alignment meetings in an **Agile environment**, surfacing blockers and aligning on sprint priorities to consistently ship features on schedule

Software Developer Intern

January 2026 – April 2026

Emergconnect

Waterloo, ON, Canada

- **Optimized an NLP classification pipeline**, reducing patient complaint **misclassification rate by 25% across 5 triage categories**, by refactoring model logic and implementing systematic **prompt refinement and error analysis**
- **Built automated evaluation pipelines** using **test case design and confusion matrix analysis**, identifying edge cases across 200+ patient samples and refining classification thresholds to improve **prediction consistency by 30%**
- **Documented** and presented findings reports to the project manager detailing model performance metrics and optimization outcomes, directly influencing the team's decision to adopt the refined NLP pipeline

Instructional Assistant

September 2025 – Present

Wilfrid Laurier University

Waterloo, ON, Canada

- **Supported 60+ students** in applying data cleaning, hypothesis testing, regression, and ANOVA in R across **4+ assignment cycles**, improving assignment accuracy by **25%** and deepening statistical understanding
- **Debugged** student R code and guided data visualization using ggplot2 and Matplotlib across **200+ support interactions**, mentoring **15+ students** on ML models and reducing project error rates by **30%**
- **Developed 5+ structured** Python and R coding walkthroughs delivered through office hours and tutorials, improving student project outcomes by **20%** and boosting assignment completion rates by **35%**

PROJECTS

🔗GreenLens AI - 1st Place Data Mind Challenge | Python, Pandas, Matplotlib, ReportLab

March 2026

- **Engineered** an AI sustainability analytics platform that processed enterprise token usage data across models and departments, converting raw metrics into **cost, carbon, and water** impact calculations
- **Generated** automated PDF reports using **Python and ReportLab**, transforming raw AI usage data into **KPI summaries** and business recommendations, enabling non-technical stakeholders to make data-driven AI governance decisions
- **Processed** and cleaned enterprise AI usage data using Pandas, calculating cost, carbon, and water impact metrics across model types and departments, transforming raw token activity into targeted optimization recommendations

🔗A.U.R.A. - Winner of Bearhacks | API, Cloud Storage, Three.js, Vultr

April 2026

- **Engineered a three-agent AI pipeline** (Geometry Extraction, Placement Planning, Scene Assembly) using **Backboard APIs**, transforming 2D floor plans into structured data and fully interactive 3D environments
- **Architected and deployed a full-stack application** on **Vultr** (VPS, Object Storage, PostgreSQL, Load Balancer), enabling **scalable scene generation, persistent storage, and real-time job tracking** for end-to-end workflows
- **Collaborated** in a **cross-functional team** using GitHub to design intuitive user experiences and align AI outputs with real-world spatial needs, ensuring the product was robust and practically usable for diverse use cases

🔗CareSync - Winner HackCanada | Auth0, JavaScript, Node.js, React

March 2026

- **Built an AI-powered care coordination platform** by implementing **persistent patient memory** and automated briefing generation, streamlining communication between PSWs, coordinators, and families
- **Developed a multi-agent system** where automated agents detect **medication discrepancies** through nightly monitoring, generate caregiver handoff briefings, and draft personalized family updates from clinical notes
- **Implemented secure authentication and access control** to protect sensitive healthcare data

TECHNICAL SKILLS AND CERTIFICATIONS

Achievements: 6X Hackathon winner, Harvard CS50

Languages: C/C++, CSS, HTML, Java, JavaScript/TypeScript, Python, R, SQL (PostgreSQL)

Frameworks & Libraries: Flask, FastAPI, Next.js, Node.js, NumPy, Pandas, React, scikit-learn, Three.js, Vue.js

Tools: AWS, Azure, Copilot, Docker, Firebase, GitHub, Google Cloud, Jupyter Notebook, MongoDB, MySQL, Vercel, VS Code